

Problem

A “general” first-order filter is shown in Fig. 14.93.

(a) Show that the transfer function is

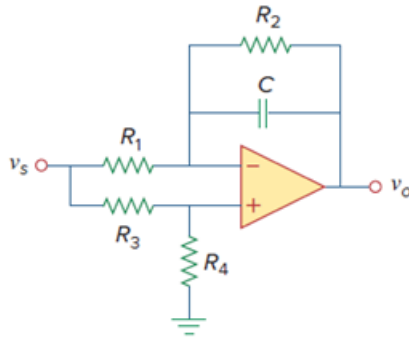
$$\mathbf{H}(s) = \frac{R_4}{R_3 + R_4} \times \frac{s + (1/R_1 C)[R_1/R_2 - R_3/R_4]}{s + 1/R_2 C},$$

$$s = j\omega$$

(b) What condition must be satisfied for the circuit to operate as a high-pass filter?

(c) What condition must be satisfied for the circuit to operate as a low-pass filter?

Figure 14.93:



Step-by-step solution

Step 1 of 9

(a)

Refer to Figure 14.93, in the textbook.

Apply Laplace transform to the components of the circuit.

The impedance of the capacitor C obtained after applying Laplace transform is $\frac{1}{Cs}$.

The feedback impedance Z_f is the parallel combination of the capacitive impedance and the resistor, R_2 .

$$\begin{aligned} Z_f &= \frac{1}{Cs} \parallel R_2 \\ &= \frac{R_2 \cdot \frac{1}{Cs}}{R_2 + \frac{1}{Cs}} \\ &= \frac{R_2}{1 + R_2 Cs} \end{aligned}$$

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Apply superposition theorem and determine the output for input at both inverting and non-inverting terminal, individually.

First ground the non-inverting terminal and calculate the output due to input at the inverting terminal.

The output of an inverting amplifier for an input of $V_s(s)$ is,

$$V_{oi}(s) = -\frac{Z_f}{Z_1} V_s(s)$$

Substitute $\frac{R_2}{1+R_2Cs}$ for Z_f and R_1 for Z_1 .

$$\begin{aligned} V_{oi}(s) &= -\frac{\frac{R_2}{1+R_2Cs}}{R_1} V_s(s) \\ &= -\frac{R_2}{R_1(1+R_2Cs)} V_s(s) \end{aligned}$$

Step 3 of 9

Calculate the output due to input at non-inverting terminal; short circuit the inverting terminal.

Apply voltage division rule to determine the voltage at the non-inverting terminal.

The

$$V_{i(NI)}(s) = \frac{R_4}{R_3 + R_4} V_s(s)$$

The output voltage of a non-inverting amplifier due to a voltage $V_{i(NI)}(s)$ is,

$$V_{oni}(s) = \left(1 + \frac{Z_f}{Z_1}\right) V_{i(NI)}(s)$$

Substitute $\frac{R_2}{1+R_2Cs}$ for Z_f , R_1 for Z_1 and $\frac{R_4}{R_3 + R_4} V_s(s)$ for $V_{i(NI)}(s)$.

$$\begin{aligned} V_{oni}(s) &= \left(1 + \frac{\frac{R_2}{1+R_2Cs}}{R_1}\right) \frac{R_4}{R_3 + R_4} V_s(s) \\ &= \left(1 + \frac{R_2}{R_1(1+R_2Cs)}\right) \frac{R_4}{R_3 + R_4} V_s(s) \end{aligned}$$

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The total output voltage is the sum of output voltages due to inputs at inverting and non-inverting terminals taken individually.

$$V_o(s) = V_{oi}(s) + V_{oni}(s)$$

Substitute $\left(1 + \frac{R_2}{R_1(1 + R_2Cs)}\right) \frac{R_4}{R_3 + R_4} V_s(s)$ for $V_{oni}(s)$ and $-\frac{R_2}{R_1(1 + R_2Cs)} V_s(s)$ for $V_{oi}(s)$.

$$\begin{aligned} V_o(s) &= -\frac{R_2}{R_1(1 + R_2Cs)} V_s + \left(1 + \frac{R_2}{R_1(1 + R_2Cs)}\right) \frac{R_4}{R_3 + R_4} V_s(s) \\ &= \frac{R_4}{R_3 + R_4} \left(1 + \frac{R_2}{R_1(1 + R_2Cs)} - \frac{R_2(R_3 + R_4)}{R_1R_4(1 + R_2Cs)}\right) V_s(s) \\ &= \frac{R_4}{R_3 + R_4} \left(\frac{R_1R_4(1 + R_2Cs) + R_4 - R_2(R_3 + R_4)}{R_1R_4(1 + R_2Cs)}\right) V_s(s) \\ &= \frac{R_4}{(R_3 + R_4)(1 + R_2Cs)} \left(1 + R_2Cs + \frac{1}{R_1} - \frac{R_2R_3}{R_1R_4} - \frac{1}{R_1}\right) V_s(s) \end{aligned}$$

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Simplify further to obtain the total output voltage.

$$\begin{aligned} V_o(s) &= \frac{R_4}{(R_3 + R_4) \left(\frac{1}{R_2C} + s\right)} \left(\left(\frac{1}{R_2C} + s\right) - \frac{R_2R_3}{R_1R_4R_2C}\right) V_s(s) \\ &= \frac{R_4}{(R_3 + R_4) \left(\frac{1}{R_2C} + s\right)} \left(s + \frac{1}{R_2C} - \frac{R_3}{R_1R_4C}\right) V_s(s) \\ &= \frac{R_4}{(R_3 + R_4) \left(\frac{1}{R_2C} + s\right)} \left(s + \frac{1}{R_1C} \left(\frac{R_1}{R_2} - \frac{R_3}{R_4}\right)\right) V_s(s) \end{aligned}$$

The transfer function $H(s)$ of a circuit is given by the ratio of the output to the input.

$$\begin{aligned} H(s) &= \frac{V_o(s)}{V_s(s)} \\ &= \frac{R_4}{(R_3 + R_4) \left(\frac{1}{R_2C} + s\right)} \left(s + \frac{1}{R_1C} \left(\frac{R_1}{R_2} - \frac{R_3}{R_4}\right)\right) \end{aligned}$$

Therefore, the transfer function is,

$$\boxed{H(s) = \frac{R_4}{(R_3 + R_4) \left(\frac{1}{R_2C} + s\right)} \left(s + \frac{1}{R_1C} \left(\frac{R_1}{R_2} - \frac{R_3}{R_4}\right)\right)}$$

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(b)

The magnitude of the transfer function must be one for $\omega = \infty$, for the circuit to function as an ideal high pass filter.

$$H(j\omega) = \frac{R_4}{(R_3 + R_4) \left(\frac{1}{j\omega} \cdot \frac{1}{R_2 C} + 1 \right)} \left(1 + \frac{1}{j\omega} \cdot \frac{1}{R_1 C} \left(\frac{R_1}{R_2} - \frac{R_3}{R_4} \right) \right)$$

Substitute ∞ for ω .

$$\begin{aligned} H(\infty) &= \frac{R_4}{(R_3 + R_4) \left(\frac{1}{\infty} \cdot \frac{1}{R_2 C} + 1 \right)} \left(1 + \frac{1}{\infty} \cdot \frac{1}{R_1 C} \left(\frac{R_1}{R_2} - \frac{R_3}{R_4} \right) \right) \\ &= \frac{R_4}{(R_3 + R_4)(0+1)} \left(1 + 0 \cdot \frac{1}{R_1 C} \left(\frac{R_1}{R_2} - \frac{R_3}{R_4} \right) \right) \quad \left(\text{Since } \frac{1}{\infty} = 0 \right) \\ &= \frac{R_4}{R_3 + R_4} \end{aligned}$$

The requisite filter is a practical filter; so, it attenuates the signal but still passes those frequencies.

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The magnitude of the transfer function must be zero for $\omega = 0$.

Substitute $s = j\omega$ in the expression for transfer function.

$$H(\omega) = \frac{R_4}{(R_3 + R_4) \left(\frac{1}{R_2 C} + j\omega \right)} \left(j\omega + \frac{1}{R_1 C} \left(\frac{R_1}{R_2} - \frac{R_3}{R_4} \right) \right)$$

Substitute 0 for $j\omega$.

$$\begin{aligned} H(0) &= \frac{R_4}{(R_3 + R_4) \left(\frac{1}{R_2 C} + 0 \right)} \left(0 + \frac{1}{R_1 C} \left(\frac{R_1}{R_2} - \frac{R_3}{R_4} \right) \right) \\ &= \frac{R_4}{(R_3 + R_4) \left(\frac{1}{R_2 C} \right)} \left(\frac{1}{R_1 C} \left(\frac{R_1}{R_2} - \frac{R_3}{R_4} \right) \right) \\ &= \frac{R_4}{(R_3 + R_4) \left(\frac{1}{R_2 C} \right)} \left(\frac{1}{R_2 C} - \frac{R_3}{R_4 R_1 C} \right) \\ &= \frac{R_4 R_2 C}{(R_3 + R_4)} \left(\frac{R_4 R_1 - R_3 R_2}{R_4 R_2 R_1 C} \right) \end{aligned}$$

For a high pass filter $H(0) = 0$.

$$0 = \frac{R_4 R_2 C}{(R_3 + R_4)} \left(\frac{R_4 R_1 - R_3 R_2}{R_4 R_2 R_1 C} \right)$$

$$R_4 R_1 - R_3 R_2 = 0$$

$$R_4 R_1 = R_3 R_2$$

Thus, the condition for the given filter to function as a high pass filter is, $\boxed{R_4 R_1 = R_3 R_2}$.

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(c)

The magnitude of the transfer function must be zero for $\omega = \infty$, for the circuit to function as an ideal low pass filter.

$$H(j\omega) = \frac{R_4}{(R_3 + R_4) \left(\frac{1}{j\omega} \cdot \frac{1}{R_2 C} + 1 \right)} \left(1 + \frac{1}{j\omega} \cdot \frac{1}{R_1 C} \left(\frac{R_1}{R_2} - \frac{R_3}{R_4} \right) \right)$$

Substitute ∞ for ω .

$$\begin{aligned} H(\infty) &= \frac{R_4}{(R_3 + R_4) \left(\frac{1}{\infty} \cdot \frac{1}{R_2 C} + 1 \right)} \left(1 + \frac{1}{\infty} \cdot \frac{1}{R_1 C} \left(\frac{R_1}{R_2} - \frac{R_3}{R_4} \right) \right) \\ &= \frac{R_4}{(R_3 + R_4)(0+1)} \left(1 + 0 \cdot \frac{1}{R_1 C} \left(\frac{R_1}{R_2} - \frac{R_3}{R_4} \right) \right) \quad \left(\text{Since } \frac{1}{\infty} = 0 \right) \\ &= \frac{R_4}{R_3 + R_4} \end{aligned}$$

Thus, the magnitude of the transfer function is $H(\infty) = \frac{R_4}{R_3 + R_4}$

Substitute 0 for $H(\infty)$.

$$\begin{aligned} 0 &= \frac{R_4}{R_3 + R_4} \\ R_4 &= 0 \end{aligned}$$

But R_4 "cannot be zero" as this condition leads to an all pass filter.

Hence, $R_4 \neq 0$, it has to have a finite value for the filter to function as a low pass filter.

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The magnitude of the transfer function must be one for $\omega = 0$.

Substitute $s = j\omega$ in the expression for transfer function.

$$H(\omega) = \frac{R_4}{(R_3 + R_4) \left(\frac{1}{R_2 C} + j\omega \right)} \left(j\omega + \frac{1}{R_1 C} \left(\frac{R_1}{R_2} - \frac{R_3}{R_4} \right) \right)$$

Substitute 0 for $j\omega$.

$$\begin{aligned} H(0) &= \frac{R_4}{(R_3 + R_4) \left(\frac{1}{R_2 C} + 0 \right)} \left(0 + \frac{1}{R_1 C} \left(\frac{R_1}{R_2} - \frac{R_3}{R_4} \right) \right) \\ &= \frac{R_4}{(R_3 + R_4) \left(\frac{1}{R_2 C} \right)} \left(\frac{1}{R_1 C} \left(\frac{R_1}{R_2} - \frac{R_3}{R_4} \right) \right) \\ &= \frac{R_4}{(R_3 + R_4) \left(\frac{1}{R_2 C} \right)} \left(\frac{1}{R_2 C} - \frac{R_3}{R_4 R_1 C} \right) \\ &= \frac{R_4 R_2 C}{(R_3 + R_4)} \left(\frac{R_4 R_1 - R_3 R_2}{R_4 R_2 R_1 C} \right) \end{aligned}$$

For a low pass filter $H(0) = 1$.

$$\begin{aligned} 1 &= \frac{R_4 R_2 C}{(R_3 + R_4)} \left(\frac{R_4 R_1 - R_3 R_2}{R_4 R_2 R_1 C} \right) \\ R_4 R_1 - R_3 R_2 &= R_1 C (R_3 + R_4) \end{aligned}$$

Thus, for the filter to function as a low pass filter the following condition needs to be satisfied,

$$\boxed{R_4 R_1 - R_3 R_2 = R_1 C (R_3 + R_4)}$$

